

DEVELOPMENT OF AI-BASED EDGE DEVICE FOR NONLINEAR CONTROL STRATEGIES

A PROJECT REPORT

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ABSTRACT

This project presents the design and implementation of an **AI-based edge device for nonlinear control strategies**, aimed at improving the accuracy, adaptability, and efficiency of control systems in laboratory and industrial environments. The proposed system employs sensors and transmitters with standard 4–20 mA outputs to acquire real-time process parameters such as flow, pressure, and temperature, which are conditioned and converted into digital signals for further processing.

An embedded edge platform, specifically the STM32 Nucleo-F446RE, is utilized to execute the control algorithm locally. A trained artificial intelligence model is integrated into the system to capture nonlinear dynamics and replace conventional control approaches. The model is converted into ONNX format and deployed on the edge device, enabling real-time inference without dependency on cloud infrastructure. This ensures reduced latency, improved reliability, and enhanced data privacy.

A multi-stage processing mechanism is implemented by combining sensor data acquisition, preprocessing, and AI-based prediction to generate accurate control signals. The system continuously monitors process variables and provides real-time responses to maintain desired setpoints under varying operating conditions.

The project includes the development of a structured workflow for calibration, signal conversion, and data validation to ensure accurate system performance. Comprehensive testing is conducted under different operating scenarios, including disturbances and nonlinear variations, to evaluate system stability and response characteristics. The results demonstrate that the proposed solution significantly enhances control precision, reduces steady-state error, and adapts effectively to nonlinear system behavior.

The processed data and control actions are logged and stored for analysis, supporting system optimization and future scalability. Overall, the project provides a secure, efficient, and intelligent framework for implementing advanced control strategies using edge AI, highlighting its potential for next-generation industrial automation systems.

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LIST OF ABBREVIATIONS

S.No	ABBREVIATION	EXPANSION
1.	ESP32	Expressif System Processor

CHAPTER 1

1.1 INTRODUCTION

This chapter deals with the problem that we are facing with our existing system and the solution along with the objectives.

1.2. PROBLEM STATEMENT

Modern industrial and laboratory control systems often rely on conventional control techniques such as PID controllers, which are designed based on linear system assumptions. However, real-world processes like flow, pressure, and temperature control exhibit strong nonlinear behavior, time-varying dynamics, and external disturbances. These complexities reduce the effectiveness of traditional controllers, leading to poor performance, increased error, and instability under varying operating conditions.

Additionally, most advanced control solutions depend on cloud-based computation, which introduces latency, dependency on network connectivity, and potential data security risks. There is a need for an intelligent, real-time, and locally deployable solution that can handle nonlinearities efficiently while ensuring faster response and reliable operation.

1.2.1 ISSUES IN THE EXISTING SYSTEM

- Conventional PID controllers are not effective in handling nonlinear and time-varying system dynamics.
- Frequent manual tuning is required to maintain desired performance.
- Inability to adapt automatically to disturbances and changing operating conditions.

- Higher steady-state error and slower response time in complex systems.
- Dependence on cloud computing introduces latency and reduces real-time performance.
- Limited intelligence in traditional embedded systems for predictive decision-making.
- Poor scalability when dealing with multi-variable or complex industrial processes.

1.2.2 NEED FOR THE PROPOSED SOLUTION

- To develop an intelligent control system capable of handling nonlinear system behavior.
- To eliminate dependency on cloud processing by enabling edge-based computation.
- To achieve faster response time and reduced latency in control actions.
- To improve system accuracy, stability, and adaptability under varying conditions.
- To integrate AI models for predictive and adaptive control strategies.
- To ensure reliable and secure operation in real-time environments.
- To provide a scalable solution suitable for modern industrial automation systems.

1.2.3 PROBLEM DEFINITION

The objective of this project is to design and implement an AI-based edge device capable of performing nonlinear control in real time. The system must acquire sensor data, process it locally using an embedded platform such as the STM32 Nucleo-F446RE, and generate accurate control signals using a trained AI model.

The solution should minimize error, improve dynamic response, and adapt to system nonlinearities without relying on external computation resources. Additionally, the system must ensure efficient data handling, real-time processing, and robustness against disturbances, thereby providing an intelligent and autonomous control framework.

1.3 OBJECTIVE

The primary objective of this project is to design and implement an intelligent, real-time, and adaptive control system using an AI-based edge device that integrates sensor data acquisition, nonlinear control modeling, embedded processing, and local decision-making to enhance system accuracy, stability, and efficiency in dynamic environments.

1.3.1 SUB OBJECTIVES

- To develop a real-time data acquisition system using sensors and transmitters (4–20 mA) for accurate measurement of process parameters such as flow, pressure, and temperature.
- To implement signal conditioning and conversion techniques for transforming analog sensor outputs into digital values suitable for processing by an embedded system.
- To design and deploy an AI-based model capable of capturing nonlinear system dynamics and generating precise control actions.
- To integrate the trained AI model into an embedded edge platform, specifically the STM32 Nucleo-F446RE, for real-time inference and control.
- To develop a closed-loop control mechanism that continuously monitors process variables and maintains desired setpoints under varying operating conditions.
- To reduce latency and improve system reliability by enabling local processing without dependency on cloud-based computation.
- To implement data logging and monitoring mechanisms for analyzing system

performance and supporting optimization.

- To design a structured workflow for calibration, validation, and troubleshooting of sensors, data acquisition, and control operations.
- To perform comprehensive testing under different nonlinear and disturbance conditions to ensure system accuracy, robustness, and adaptability.
- To develop a scalable system architecture that supports future enhancements such as multi-variable control and integration with IoT-based monitoring systems.

1.4 THRUST AREA

Artificial Intelligence and Machine Learning: The project emphasizes the use of AI models to handle nonlinear system dynamics, enabling adaptive and intelligent control beyond traditional methods.

Edge Computing and Real-Time Processing: The system focuses on local data processing using embedded platforms, reducing latency and enabling fast, real-time decision-making without cloud dependency.

Embedded Systems and Industrial Automation: The project utilizes microcontrollers, sensors, and signal conditioning circuits, contributing to advancements in embedded control systems for real-world industrial applications.

Nonlinear Control Systems: The work addresses challenges in controlling nonlinear and time-varying processes, supporting the development of advanced control strategies for complex systems.

IoT and Smart Monitoring Systems: The integration of sensors and data logging supports intelligent monitoring and forms a foundation for future IoT-based industrial automation.

1.5 SOCIAL RELEVANCE.

Improving Industrial Efficiency: The system enhances process control accuracy and reduces errors, leading to improved productivity and operational efficiency in industries.

Reducing Human Intervention: Automation of control processes minimizes manual tuning and monitoring, reducing human effort and operational complexity.

Promoting

Advanced

Technology

Adoption:

The project encourages the use of AI and edge computing in real-world applications, contributing to technological advancement and innovation.

1.6 CONCLUSION

In this chapter the detailed view about the project objective, Problem statement and social Relevance were discussed.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

This chapter deals with the Literature survey for this project.

2.2 LITERATURE SURVEY

2.2.1 PROPOSED METHODOLOGY

Proposed Methodology presents the design of an AI-based Edge Device for Nonlinear Control Strategies, aimed at improving control accuracy, adaptability, and real-time performance in nonlinear systems. The system utilizes artificial intelligence to model complex system behavior and generate optimal control actions. Real-time process data is collected and preprocessed to ensure reliable input for the AI model. The trained model learns nonlinear relationships between inputs and outputs and acts as an intelligent controller. A feedback mechanism compares the system output with the desired setpoint to enable continuous correction. The proposed approach ensures efficient handling of nonlinearities, faster response, and improved system stability, making it suitable for modern industrial control applications.

2.2.2 EXISTING METHODOLOGY

- 1 Patel S., Mehta D. “Adaptive Control Using Machine Learning Techniques.” The authors highlight the use of machine learning algorithms for adaptive control in nonlinear environments. The system continuously updates control actions based on feedback, leading to improved robustness and performance..

- 2 Chen L., Zhao Y. “Recurrent Neural Networks for Nonlinear Process Control.” This paper discusses the application of RNN models in capturing time-dependent nonlinear behavior in dynamic systems. The proposed method improves prediction accuracy and ensures better handling of system disturbances.
- 3 Smith John, Lee Wong. “Neural Network-Based Nonlinear Control for Industrial Systems.” This work presents the use of artificial neural networks to model nonlinear system dynamics and improve control accuracy. The system learns complex input-output relationships and adapts to varying operating conditions, resulting in enhanced stability and reduced control error.
- 4 Kumar R., Sharma P. “Edge AI for Real-Time Control Applications.” This study explores the integration of artificial intelligence with edge computing for low-latency control systems. The approach emphasizes real-time data processing, reduced dependency on cloud infrastructure, and improved responsiveness in industrial automation.
- 5 Brown T., Wilson G. “Implementation of AI-Based Controllers in Embedded Systems.” This work focuses on deploying AI models in embedded platforms for real-time control. It emphasizes efficient model optimization and reliable performance in resource-constrained environments.

2.3 CONCLUSION:

In this chapter the review about the literature survey for the proposed project has been discussed

CHAPTER 3

COMPONENTS DESCRIPTION AND SYSTEM INTEGRATION

3.1 INTRODUCTION

This chapter deals with the features and usage of each component which are used in this project along with the advantages.

3.2 MICROCONTROLLER -ESP32

The ESP32 development board is a powerful and versatile microcontroller used in our system as the central processing and communication unit. It handles sensor data collection, local processing, and secure wireless communication. With its dual-core processor, built-in Wi-Fi and Bluetooth, and extensive GPIO support, the ESP32 is well-suited for advanced IoT applications requiring speed, reliability, and real-time connectivity.

High-Speed Data Processing: The ESP32's dual-core 32-bit processor enables fast and efficient computation, allowing real-time processing of sensor data, authentication tasks, and system control operations.

Peripheral Expansion: With its ADCs, DACs, GPIOs, I²C, SPI, UART, and PWM support, the ESP32 interfaces easily with fingerprint sensors, cameras (OV2640), RFID modules, displays, and intrusion detection modules.

Real-Time Event Handling: The dual-core architecture allows one core to handle sensor tasks while the other manages connectivity, improving responsiveness and system stability.

3.2.1 ADVANTAGES OF USING ESP32

High Performance: The dual-core Tensilica processor provides efficient computation capability, supporting real-time data acquisition, preprocessing, and handling of multiple tasks simultaneously in control applications.

3.3 MICROCONTROLLER - NUCLEO-F446RE STM32

The NUCLEO-F446RE STM32 development board is a high-performance microcontroller platform used in our system for implementing AI-based control strategies and real-time processing. It acts as the core computational unit for executing control algorithms and handling system-level operations. Based on the ARM Cortex-M4 architecture, it provides high processing capability, making it suitable for embedded and edge AI applications requiring precision and reliability.

High-Speed Data Processing: The STM32F446RE features a 32-bit ARM Cortex-M4 processor with floating-point unit (FPU), enabling fast execution of complex mathematical computations and control algorithms required for nonlinear systems.

AI Model Execution Support: The board supports deployment of optimized AI models using frameworks such as STM32Cube.AI, allowing real-time inference and intelligent decision-making at the edge.

Real-Time Control Capability: Designed for deterministic performance, the microcontroller ensures precise timing and reliable execution of control tasks, which is essential for maintaining system stability.

3.3.1 ADVANTAGES OF USING NUCLEO-F446RE STM32

- **High Performance Processing:** The STM32F446RE is based on a 32-bit ARM Cortex-M4 core with a floating-point unit (FPU), enabling fast and efficient execution of complex mathematical operations required for nonlinear control strategies.
- **Real-Time Control Capability:** The microcontroller provides deterministic timing and high-speed execution, making it suitable for real-time control applications where precise response is critical.
- **AI Integration Support:** The board supports deployment of AI models using tools like STM32Cube.AI, allowing edge-level inference and intelligent control without dependence on external systems.
- **Rich Peripheral Interfaces:** It includes multiple communication interfaces such as ADC,

DAC, UART, SPI, I²C, and PWM, ensuring easy integration with sensors, actuators, and external modules.

- **High ADC Resolution:** The built-in Analog-to-Digital Converter offers accurate signal conversion, improving the precision of sensor data acquisition and overall system performance.
- **Low Power Consumption:** Designed for embedded applications, the STM32 ensures efficient power usage, making it suitable for continuous and long-duration operations.
- **Development Flexibility:** The NUCLEO board supports Arduino headers and STM32 development tools, enabling easy prototyping, debugging, and scalability.
- **Reliable and Robust Design:** The STM32 platform is widely used in industrial applications due to its stability, reliability, and long-term performance.

STM32 Nucleo-64 boards

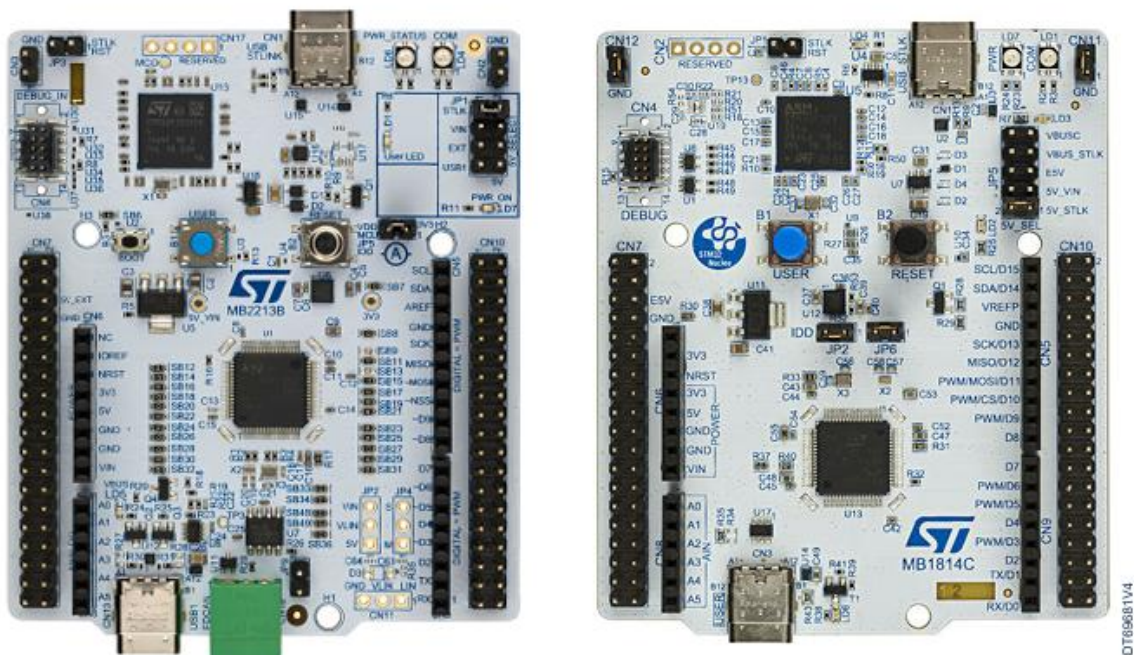


FIG 3.2. PICTURE OF NUCLEO-F446RE STM32

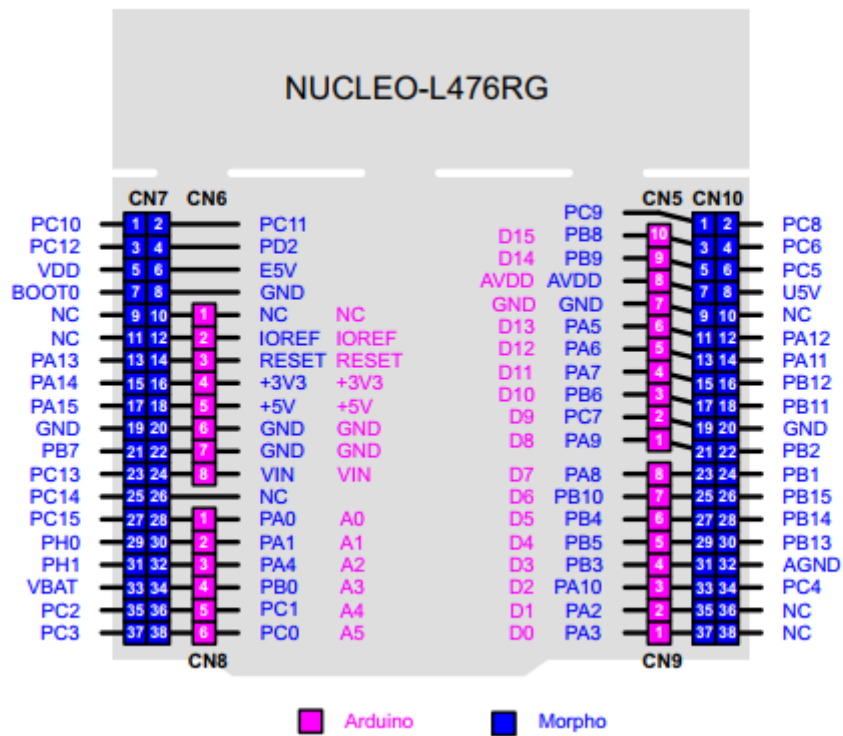


FIG 3.3 PINOUT DIAGRAM OF NUCLEO-F446RE STM32

3.4. TRANSMITTERS

3.4.1 TEMPERATURE TRANSMITTER- RESISTANCE

TEMPERATURE DETECTOR:

The RTD-based temperature transmitter is used in our system for accurate measurement and transmission of temperature data during the data acquisition process. It converts the temperature sensed by the Resistance Temperature Detector (RTD) into a standardized analog signal, typically in the range of 4–20 mA, which can be easily interfaced with embedded systems. Known for its high accuracy and stability, the RTD transmitter ensures reliable temperature monitoring in dynamic industrial environments.

3.4.1.1 ADVANTAGES OF USING RTD

High Accuracy: RTDs provide highly accurate temperature measurements with minimal error, making them suitable for precise data acquisition in control systems.

Excellent Linearity: The resistance of RTD changes almost linearly with temperature, simplifying calibration and improving measurement reliability.

High Stability: RTDs offer superior long-term stability and repeatability, ensuring consistent performance over extended periods of operation.

Wide Temperature Range: Capable of measuring a broad range of temperatures, making them versatile for various industrial and experimental applications.

Noise-Resistant Transmission: The 4–20 mA output signal is less affected by electrical noise and signal loss, ensuring reliable data transmission over long distances.

Robust and Durable: RTDs are designed to withstand harsh industrial environments, providing reliable operation under varying conditions.

Easy Integration: The standardized output can be easily interfaced with microcontrollers such as ESP32 and STM32 using simple signal conditioning circuits.

Low Drift: RTDs exhibit minimal drift over time, maintaining accuracy without frequent recalibration.



FIG 3.5 TEMPERATURE TRANSMITTER AT FLOWLAB

3.4.1.2 SPECIFICATIONS OF RTD

Parameter	Specification
Sensor Type	Pt100 / Pt1000
Temperature Range	-196°C to +600°C
Measuring Current	0.1 mA to 1.0 mA
Accuracy Class B	$\pm(0.30 + 0.0050)$
Accuracy Class A	$\pm(0.15 + 0.0020)$
Accuracy Class AA	$\pm(0.10 + 0.0017)$
Wiring Configuration	2-wire / 3-wire / 4-wire
Ingress Protection	Up to IP65 / IP66
Probe Diameter	3 mm, 6 mm, 8 mm
Spring Travel	Max. 10 mm



FIG 3.6 RESISTANCE TEMPERATURE DETECTOR

3.4 .2 DIFFERENTIAL PRESSURE FLOWMETER:

The Differential Pressure (DPT) flowmeter is used in our system to measure the flow rate of fluids by utilizing the pressure difference generated across a flow restriction element such as an orifice plate or venturi tube. It operates based on Bernoulli's principle, where the fluid velocity increases and pressure decreases at the restriction point. The pressure difference is measured by a differential pressure transmitter and is used to calculate the flow rate. Due to its simplicity, reliability, and wide industrial usage, the DPT flowmeter is well-suited for process monitoring and control applications.

Working Principle: The flowmeter measures the pressure difference between two points in a pipeline. This differential pressure is proportional to the square of the flow rate, enabling flow calculation using standard equations.

High Reliability: DPT flowmeters have a simple design with no moving parts, resulting in low maintenance and long operational life.

Wide Application Range: Suitable for measuring flow of liquids, gases, and steam in various industrial environments.

Cost-Effective Solution: Compared to advanced flowmeters, DPT-based systems are economical while providing acceptable accuracy for most applications.

Standard Industrial Output: The differential pressure transmitter provides a 4–20 mA output signal, which can be easily interfaced with controllers and embedded systems.

Easy Integration: Can be integrated with microcontrollers like ESP32 and STM32 for real-time data acquisition and control.

3.4.2.1 ADVANTAGES OF USING DPT:

Simple and Robust Design: The DPT flowmeter has no moving parts, making it highly reliable and reducing chances of mechanical failure.

Cost-Effective: It is more economical compared to advanced flow measurement techniques, making it suitable for industrial and academic applications.

Wide Applicability: Can be used for measuring the flow of liquids, gases, and steam across a wide range of operating conditions.

Low Maintenance: Due to its simple construction, it requires minimal maintenance and has a long service life.

Good Accuracy: Provides acceptable accuracy for most process control applications when properly calibrated.

Standard Output Signal: Generates a 4–20 mA output, which is easy to interface with controllers, PLCs, and microcontrollers.

Easy Installation: Can be easily installed using standard primary elements like orifice plates, venturi tubes, or flow nozzles.

Suitable for Harsh Environments: Performs well in high temperature and pressure conditions, making it ideal for industrial use.



FIG 3.7 Orifice meter at Flow lab

3.4.2.1 SPECIFICATIONS OF DIFFERENTIAL PRESSURE TRANSMITTER

Parameter	Specification
Output Signal	4–20 mA DC
Low Power Output	1–5 V
Communication Protocol	Hart
Supply Voltage	Up to 42 V DC
Ambient Temperature	–40°C to 60°C
Storage Temperature	–40°C to 85°C
Humidity Range	0–100% RH
Process Temperature	Up to 120°C
Measurement Types	Differential, Gauge, Absolute Pressure
Wiring Configuration	2-wire loop powered system
Mounting	Bracket / Pipe / Wall mounting options
Explosion Protection	Intrinsically Safe



FIG 3.8 Flow Transmitter at Flow lab

3.5 CONCLUSION:

In this chapter the detailed description about the components used along with its specifications and advantages, how they are used in this proposed project has been discussed

CHAPTER 4

REAL-TIME DATA ACQUISITION USING ESP32 FROM INDUSTRIAL TRANSMITTERS

4.1 INTRODUCTION

In this chapter, we shall discuss the data acquisition process using the ESP32 from the temperature transmitter and Differential Pressure (DPT) flow transmitter. The system is designed to collect process data for monitoring, analysis, and development of AI-based control strategies. The overall data collection process is carried out in two phases: synthetic data generation and real-time data acquisition.

4.1.1 PHASE 1: SYNTHETIC DATA GENERATION USING MATLAB

In the initial phase, synthetic data is generated using MATLAB to simulate the behavior of temperature and flow parameters. This phase is essential when real-time data is not readily available during the early stages of system development.

MATLAB is used to create datasets based on mathematical models and assumed operating conditions. The generated data mimics real industrial signals, including variations in temperature and flow rate over time. This synthetic dataset includes:

Temperature (°C)

Flow Rate

Time Stamp

The synthetic data is used for preliminary analysis, testing, and training of AI models. It helps in validating system performance and ensures a smooth transition to real-time implementation.

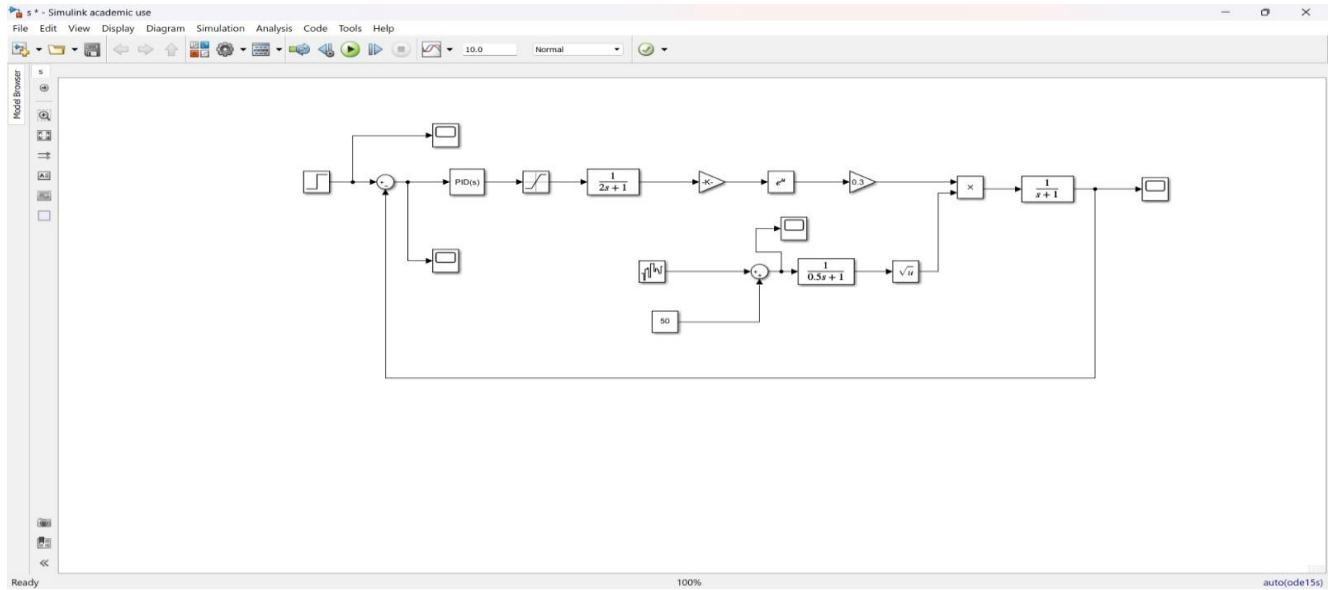


FIG 4.1 MATLAB Simulink Model for Synthetic Data Generation and Collection

4.1.2 PHASE 2: REAL-TIME DATA COLLECTION USING ESP32

Overview of Data Collection System

In this phase, real-time data is collected using the ESP32 from temperature and DPT flow transmitters. The system is designed to acquire accurate process parameters for monitoring and control applications.

Temperature Data Acquisition

The temperature of the process is measured using an RTD-based temperature transmitter. The transmitter converts the sensed temperature into a standardized 4–20 mA current signal, where 4 mA represents the minimum temperature and 20 mA represents the maximum temperature range.

Flow Data Acquisition Using DPT

The flow rate is measured using a Differential Pressure (DPT) transmitter. It measures the pressure difference across a flow element (such as an orifice plate) and generates a 4–20 mA output signal proportional to the flow rate.

Signal Conditioning (4–20 mA to Voltage Conversion)

Since the ESP32 accepts voltage input, the current signals from both transmitters are converted into voltage signals. A **250 Ω precision resistor** is used to convert the 4–20

mA signal into a **1–5 V range**. This is then scaled appropriately (using voltage division if required) to match the ESP32 ADC input range (0–3.3 V).

Data Acquisition Using ESP32

The ESP32 reads the conditioned analog signals through its ADC channels. These analog values are converted into digital form and scaled to obtain actual physical parameters such as temperature and flow rate.

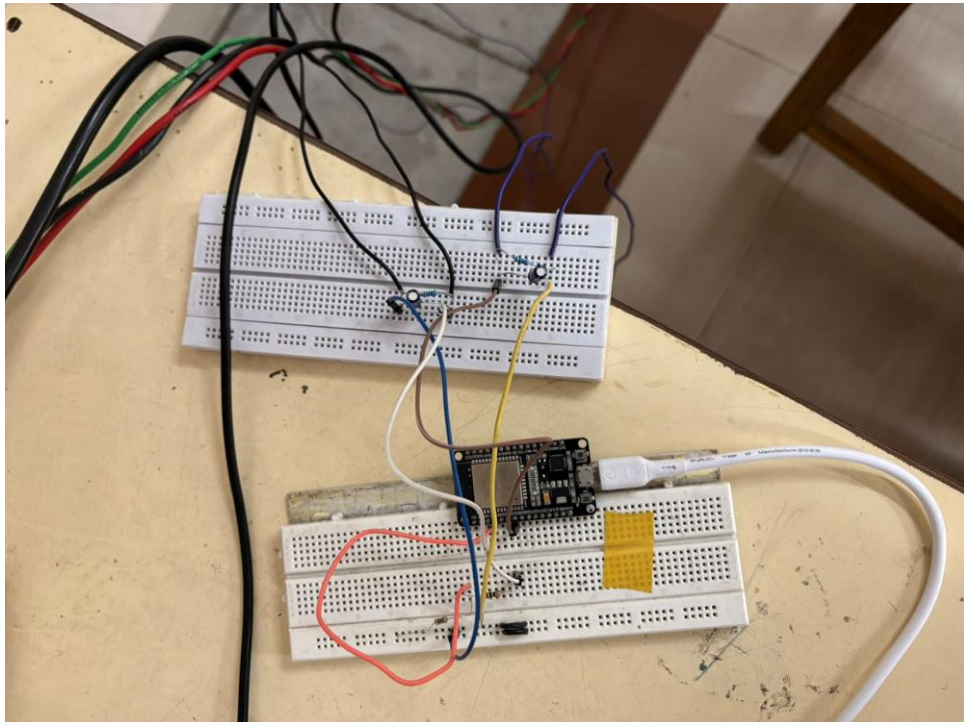


FIG 4.2 Hardware Setup for ESP32-Based Data Collection System

Serial Communication and Data Logging

The processed data is transmitted from the ESP32 to a computer using serial communication (UART). A Python-based program reads the incoming data, assigns timestamps, and stores it in a CSV file for further analysis and model development.

Collected Parameters

The system collects the following parameters:

1. Temperature (°C)
2. Flow Rate (from DPT)
3. Time Stamp

Time	Flow(mmH2O)	Temperature(°C)
16:38:22	160.989	29.253
16:38:23	306.447	28.789
16:38:24	418.462	29.011
16:38:25	637.656	29.051
16:38:26	777.07	28.709
16:38:27	902.381	29.253
16:38:28	840.733	29.051
16:38:29	878.205	28.91
16:38:30	928.571	29.112
16:38:31	1014.799	29.152
16:38:32	1089.744	29.152
16:38:33	1161.062	28.628
16:38:34	1309.341	29.031
16:38:35	1290.403	29.353
16:38:36	1311.355	30.24
16:38:37	1200.952	29.253
16:38:38	1400	29.253
16:38:39	1286.374	29.233
16:38:40	1152.601	29.172
16:38:41	1320.623	29.253
16:38:42	1202.564	29.233
16:38:43	1050.256	29.192
16:38:44	1035.348	29.253
16:38:45	930.183	29.353
16:38:46	826.227	29.374
16:38:47	781.099	29.031

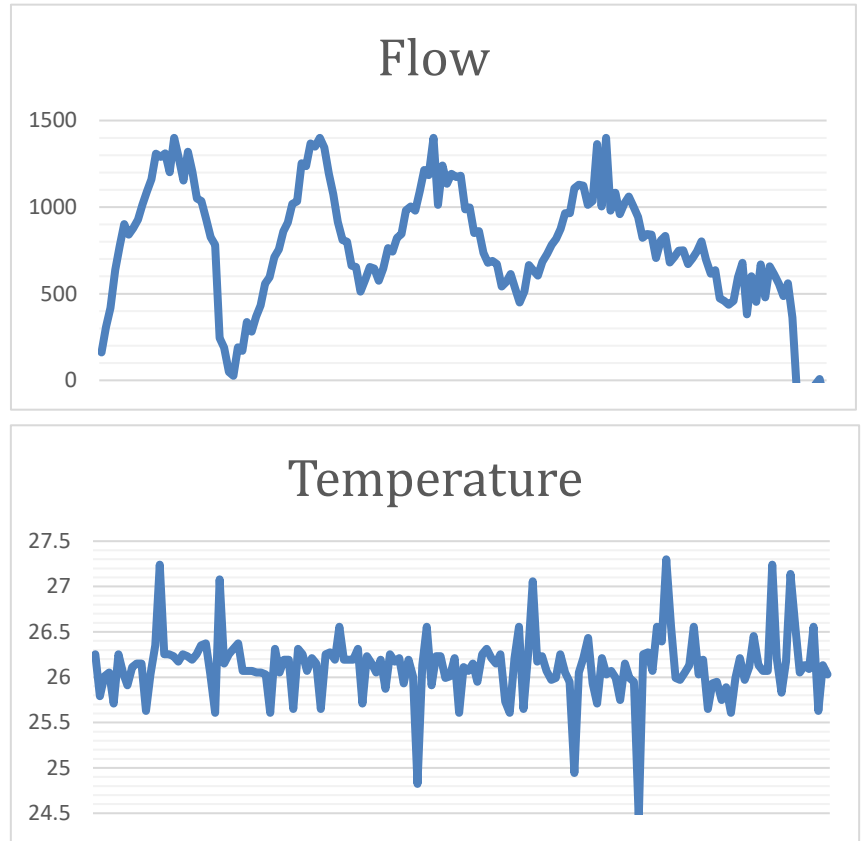


FIG 4.3 Collected data

4.2 CONCLUSION

The proposed two-phase approach ensures efficient system development. The synthetic data phase enables initial testing and model training, while the real-time data acquisition phase provides accurate and continuous process data. The use of ESP32 ensures a cost-effective, reliable, and real-time data acquisition system, forming a strong foundation for implementing AI-based nonlinear control strategies.

CHAPTER 5

MODEL TRAINING AND EDGE DEPLOYMENT

5.1 INTRODUCTION

This phase focuses on transforming the acquired process data into an intelligent predictive control system. The collected real-time data from the ESP32-based acquisition setup is used to train a deep learning model capable of predicting flow behavior. A Gated Recurrent Unit (GRU) based model is developed due to its effectiveness in handling time-series data.

The trained model is then deployed onto an embedded edge device, STM32 Nucleo-F446RE, enabling real-time prediction and control signal generation to maintain desired flow conditions.

5.2 DATA PREPARATION AND PREPROCESSING

The raw data collected from sensors (such as flow, temperature, and pressure transmitters) is first stored in CSV format and processed using Python in the PyCharm environment.

Key preprocessing steps include:

- **Data Cleaning:**

Removal of noise, invalid readings, and missing values to ensure data integrity.

- **Normalization:**

The input features are scaled (typically between 0 and 1) to improve model convergence and stability.

- **Time-Series Structuring:**

Since flow behavior depends on past values, the dataset is converted into sequential format using sliding windows.

Feature Selection:

Relevant parameters such as pressure, temperature, and previous flow values are selected as inputs to the model

5.3 GRU-BASED MODEL DEVELOPMENT

A **Gated Recurrent Unit (GRU)** neural network is chosen for this application due to its ability to capture temporal dependencies in sequential data while maintaining computational efficiency.

Model characteristics:

- **Input Layer:**
Accepts time-series sequences of sensor data.
- **GRU Layers:**
Extract temporal patterns and relationships in flow dynamics.
- **Dense Output Layer:**
Produces the predicted flow value.

5.4 MODEL TRAINING IN PYCHARM

The model is implemented and trained using Python in the PyCharm environment with the support of deep learning and data processing libraries such as:

- TensorFlow / Keras
- **PyTorch**
- NumPy
- Pandas

Both TensorFlow/Keras and PyTorch frameworks are suitable for developing deep learning models; in this work, PyTorch is utilized for its flexibility, dynamic computation graph, and ease of debugging during model development.

Training Process:

- The collected dataset is divided into **training and validation sets** to evaluate model performance and avoid overfitting.
- The time-series data is fed into the GRU model in sequential format.

- The model is trained using **Backpropagation Through Time (BPTT)** to capture temporal dependencies in the data.
- **Loss Function:** Mean Squared Error (MSE) is used to measure prediction accuracy.
- **Optimizer:** Adam optimizer is employed for efficient and adaptive learning.
- During training, model parameters are iteratively updated to minimize prediction error.

Training Outcomes:

- The model learns the complex relationship between process variables (such as pressure, temperature, and previous flow values) and the flow rate.
- It achieves accurate prediction of flow under varying operating conditions.
- The training loss decreases over epochs, indicating improved model performance and convergence.

5.5 MODEL CONVERSION FOR EMBEDDED DEPLOYMENT

Since the trained model cannot be directly deployed on microcontrollers, it is converted into an optimized format.

Conversion steps:

1. Trained model → Converted to **ONNX (Open Neural Network Exchange) format**
2. ONNX model → Optimized for embedded inference
3. Model parameters are quantized (if required) to reduce memory usage

Importance of ONNX:

- Provides interoperability between training and deployment platforms
- Enables execution on resource-constrained devices like STM32

5.6 DEPLOYMENT ON STM32 NUCLEO-F446RE

The optimized model is deployed on the **stm32 nucleo-f446re** edge device.

Deployment process:

- Model is integrated into embedded c environment
- Input data from sensors is fed into the model in real-time
- The model performs inference on-device

Advantages:

- No need for cloud computation
- Low latency real-time predictions
- Improved system reliability

5.7 NOVELTY OF THE PROPOSED SYSTEM

The proposed system introduces a novel integration of data acquisition, deep learning, and embedded edge computing for real-time industrial process control. Unlike conventional systems, which rely on static models or cloud-based computation, this work combines intelligent prediction with on-device processing.

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5.8 CONCLUSION

In this phase, the collected data was used to train a GRU-based model capable of predicting flow behavior accurately. The model was developed and validated using Python frameworks in PyCharm, ensuring effective learning of time-dependent process dynamics.

The trained model was then converted into an optimized format and successfully deployed on the STM32 Nucleo-F446RE edge device. This enabled real-time, low-latency inference directly on embedded hardware, demonstrating the feasibility of integrating machine learning with edge computing systems.

CHAPTER 6

SUMMARY AND CONCLUSION

6.1 SUMMARY

This project focuses on the **development of an AI-based edge device for implementing nonlinear control strategies**, with the primary objective of achieving accurate, real-time prediction of flow parameters in an industrial process environment. The proposed system integrates data acquisition, machine learning, and embedded deployment into a single, efficient framework.

The initial phase of the project involves real-time data acquisition using the ESP32 microcontroller. Industrial transmitters such as temperature and differential pressure sensors provide standardized 4–20 mA signals, which are converted into measurable voltage levels using appropriate signal conditioning circuits. The ESP32 reads these analog values through its ADC channels and transmits the collected data to a computer for storage and further processing. This setup ensures continuous monitoring and reliable data collection under varying operating conditions.

Following data acquisition, the collected dataset is preprocessed in the PyCharm environment using Python. This includes data cleaning, normalization, and structuring into time-series sequences suitable for model training. The preprocessed data is then used to develop a deep learning model based on the **Gated Recurrent Unit (GRU)** architecture. The GRU model is particularly suitable for this application as it efficiently captures temporal dependencies and dynamic variations in process parameters such as flow, pressure, and temperature.

The model is trained using advanced machine learning libraries such as PyTorch and TensorFlow/Keras. During training, the dataset is divided into training and validation sets to ensure proper evaluation and to avoid overfitting. The model learns the nonlinear relationships between input variables and flow rate using backpropagation through time,

optimizing its performance using the Adam optimizer and minimizing prediction error through the Mean Squared Error loss function.

Once the model achieves satisfactory performance, it is converted into an interoperable format using ONNX (Open Neural Network Exchange). This conversion enables seamless transfer of the trained model from the development environment to embedded hardware platforms. The optimized model is then deployed on the STM32 Nucleo-F446RE microcontroller, which serves as the edge device in this system.

The deployment enables real-time inference directly on the embedded device, eliminating the dependency on cloud-based computation. This significantly reduces latency, enhances system responsiveness, and improves reliability in industrial applications. The system demonstrates the capability of executing AI models on resource-constrained hardware while maintaining efficient performance.

Overall, this project successfully combines IoT-based data acquisition, deep learning-based prediction, and embedded edge computing to develop an intelligent and scalable solution for nonlinear process monitoring. It highlights the practical feasibility of integrating artificial intelligence with embedded systems for real-time industrial applications, paving the way for advanced smart control systems.

6.1 CONCLUSION

Overall, the system demonstrates how advanced digital technologies such as artificial intelligence and edge computing can be effectively leveraged to enhance industrial process monitoring and prediction. The proposed solution addresses key limitations of conventional control systems, including delayed response, lack of adaptability, and dependence on centralized computation.

Furthermore, the developed framework provides a strong foundation for future enhancements such as adaptive control strategies, integration with reinforcement learning, and large-scale deployment in smart industrial environments. The successful implementation of this project highlights the potential for developing intelligent, scalable, and real-time edge-based systems capable of improving efficiency, reliability, and performance in nonlinear process control applications.

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